

Weekly Progress Report

Project: Heart Disease Data Analysis

Week: 1

Overview

During the first week of our project, our primary focus was on collecting and preprocessing datasets relevant to heart disease. Our goal was to compile and analyze data on various health and economic indicators, ultimately merging key features into a single, well-structured CSV file.

The selected datasets covered factors such as **Alcohol Consumption, GDP per Capita, Heart Disease Mortality Rate, Heart Disease Incidence Rate, Heart Disease Prevalence, Rate of Diabetes, Prevalence of Insufficient Physical Activity, Prevalence of Obesity (BMI $\geq$ 30), Rate of Overweight, Life Expectancy, and Health Expenditure.**

To streamline the workload, we distributed dataset responsibilities among the four team members:

- **Collins:** Heart Disease Mortality Rate, Heart Disease Incidence Rate, Heart Disease Prevalence
- **Emmanuella:** Prevalence of Insufficient Physical Activity, Prevalence of Obesity, GDP per Capita
- **Nicole:** Alcohol Consumption, Rate of Diabetes
- **Prince:** Rate of Overweight, Life Expectancy, Health Expenditure

This division allowed each member to focus on cleaning, merging, and analyzing specific datasets efficiently.

Progress and Challenges of Team Members

Collins

Key Activities this Week

Data Cleaning and Preprocessing:

- Successfully loaded and cleaned three datasets (Heart Disease Mortality Rate, Incidence Rate, and Prevalence) using Google Colab.
- Removed unnecessary columns, including identifiers and redundant categorical features, to optimize dataset structure.
- Benefited from consistent column headers across the three datasets, making the cleaning process smoother.

Merging Datasets:

- Merged the datasets using **Country, Sex, and Year** as key columns, ensuring proper alignment of data points.

Country Coding:

- Applied the **pycountry** library to assign ISO Alpha-3 country codes.
- Encountered missing codes for certain countries (e.g., *Bolivia, Democratic Republic of the Congo, Taiwan, and Venezuela*), which were manually mapped using a coding dictionary.

Challenges Encountered

- **Missing Data:** Some countries and years had gaps in data, requiring **polynomial fitting imputation** to preserve dataset integrity.
- **Duplicates in Merged Data:** Redundant rows emerged due to many-to-many relationships, requiring additional filtering.
- **Large Data Size:** The merged dataset became significantly large, causing performance issues in further processing.

Conclusion

The first week was productive, as the datasets were successfully cleaned and merged. Challenges were addressed through proper imputation, filtering, and manual country mapping.

Emmanuella

Key Activities This Week

Data Collection and Preprocessing:

- Gathered data from **WHO, World Bank, and Global Burden of Disease databases**.
- Handled missing values through imputation techniques.
- Normalized and scaled variables for comparability.
- Categorized countries based on economic status for deeper analysis.

Exploratory Data Analysis (EDA):

- **Visualizations:** Used **heatmaps, histograms, and boxplots** to explore relationships and data distributions.
- **Descriptive Statistics:** Computed **mean, median, and standard deviation** for key indicators (physical activity, obesity, GDP).
- **Outlier Detection:** Found that countries with exceptionally high GDP per capita (e.g., *Luxembourg, Qatar, Bermuda*) were driven by factors such as **small populations, resource wealth, and financial sector dominance**.

Insights and Observations:

- Higher **GDP per capita correlates with higher obesity rates** but lower levels of insufficient physical activity.
- Economically disadvantaged regions often face **both physical inactivity and obesity challenges**, though these affect different population segments.

Techniques and Tools Used:

- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn
- **Imputation Methods:** Used **KNN Imputer** to handle missing values for better accuracy.
- **Correlation Analysis:** Quantified relationships between economic status and health indicators.

Conclusion

EDA revealed important trends linking economic prosperity to health outcomes. Further analysis will explore causation and refine data processing techniques.

Nicole

Progress Made

Data Preprocessing:

- Derived data from the **NCD Risk Factor Collaboration (NCD-RisC)** and the **World Bank**.
- Standardized column names across datasets (e.g., *ISO Code* → *Code*, *Country/Territory* → *Location*).
- Converted **Year** column from object to integer for uniformity across datasets.
- Removed incorrect location entries and regions to ensure data consistency.

Handling Missing Data:

- Used **regional data** to fill missing country values by mapping each country to its respective region (e.g., *Aruba* → *Caribbean small states*).
- Applied **KNN Imputer** for missing **alcohol consumption data**.
- Used **Polynomial Fitting** (NumPy) for **diabetes data** after comparing **linear, polynomial, and spline interpolation methods**.

Challenges Faced:

- **Outliers in Diabetes Data:** Were retained because they did not deviate significantly from external statistics.
- **Imputation:** Experimented with Linear, Polynomial and Spline interpolation but they failed to yield values that complemented the overall trends in the data.
- **Missing External Data:** Struggled to find consistent external sources for imputing gaps due to **differences in methodologies**.

Conclusion

Preprocessing steps were completed, and missing data was effectively handled using **region-based imputation, statistical and machine learning techniques**. Final merged dataset showed strong positive correlation between Obesity Rate and BMI, hence BMI was dropped to avoid redundancy

Prince

Key Activities for the Week

Missing Values Management:

- Identified missing values in **health expenditure and life expectancy** datasets.
- Used multiple strategies to handle missing data:

- Forward filling missing values within each country.
- Backward filling missing values.
- Removing countries with persistent missing data.
- Filtering out non-standard country codes, including regions.

Data Integration Approach:

- Merged **overweight prevalence** and **world health data** using a **left merge** on country code, country name, and year.
- Retained only essential columns for streamlined analysis.

External Data Supplementation:

- Identified countries with missing **life expectancy data** and supplemented them with additional data from **WHO datasets**.
- Filtered supplemental data to include only **total population figures**.
- Used **imputed values** to fill missing life expectancy entries.

Final Data Refinement:

- Interpolated remaining missing values for **life expectancy and health expenditure**.
- Dropped **1999, 2022, and 2023** due to data inconsistencies.

Visualization and Analysis:

- Created multiple visualizations to understand:
 - Distribution of **health expenditure**.
 - Trends in **life expectancy**.
 - Relationship between **health expenditure and life expectancy**.
 - **Overweight prevalence** by region and sex.

Key Challenges Overcome:

- **Inconsistent country naming conventions** across datasets.
- **Missing data across different years**, requiring multi-step imputation.
- **Merging datasets with different structures** while maintaining data integrity.
- **Ensuring data quality and completeness** for meaningful analysis.

Conclusion

After failed attempts to find external data, the missing values were successfully handled using advanced imputation techniques like polynomial fitting and KNN

Summary and Next Steps

In the first week, significant progress was made in **data cleaning, merging, and exploratory analysis**. Each team member successfully handled their assigned datasets, addressing challenges like missing values, outliers, and large dataset sizes.

Next steps include:

- Conducting deeper statistical analysis.
 - Developing visualizations to highlight trends in heart disease risk factors..
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